

riskier small cap stocks at 7.8% and 16.8%, respectively. Government bonds act in the opposite way, generating higher returns at 12.3% during recessions compared to 5.9% during expansions. Investment-grade corporate bonds, which have relatively little credit risk, exhibit similar behavior. In contrast, high-yield bonds are much closer to equity, and their performance is between equity and government bonds; in fact, high-yield bonds do not have any discernable difference in mean returns over recessions and expansions.

We can see a similar pattern if we look at periods of low or high growth, as measured by real GDP or consumption growth. For example, large stocks return 8.8% during periods of low real GDP growth and 5.6% during periods of low consumption growth. During periods of high real GDP growth and high consumption growth, large stock returns average 13.8% and 17.1%, respectively. Consistent with the behavior across NBER recessions and expansions, government bonds tend to do relatively well during periods of low growth, averaging 10.0% during periods of low real GDP growth compared to 3.9% during periods of high real GDP growth.

All asset returns are much more volatile during recessions or periods of low growth. For example, large stock return volatility is 23.7% during recessions compared to 14.0% during expansions. While government bonds have higher returns during recessions, their returns are also more volatile then, with a volatility of 15.5% during recessions compared to 9.3% during expansions. It is interesting to compare the volatilities of assets over the full sample to the volatilities conditional on recessions and expansions: volatility tends to be very high during bad times.

2.2. INFLATION

High inflation tends to be bad for both stocks and bonds, as Table 7.2 shows. During periods of high inflation, all assets tend to do poorly.⁵ Large stocks average 14.7% during low inflation periods and only 8.0% during periods of high inflation. The numbers for government bonds, investment grade bonds, and high yield bonds are 8.6%, 8.8%, and 9.2%, respectively, during low inflation periods and 5.4%, 5.3%, and 6.0%, respectively, during high inflation periods. It is no surprise that high inflation hurts the value of bonds: these are instruments with fixed payments, and high inflation lowers their value in real terms. It is more surprising that stocks—which are real in the sense that they represent ownership of real, productive firms—do poorly when inflation is high. We’ll take a closer look at the inflation-hedging properties of equities in chapter 8, but for now, suffice to say that high inflation is bad for both equities and bonds. Part of the long-run risk premiums for both equities and bonds represents compensation for doing badly when inflation is high. (In chapter 11, we’ll further explore the inflation hedging properties of different asset classes.)

⁵ Deflation, which is not examined here, is also a bad time when assets tend to have low returns.